A Dual-Biometric Secured Credit Card Fraud Detection Mobile Application with Merchant Intelligence



IV YEAR / VIII SEMESTER

Team Members: Parameshwaran R, Sabarirajan T, Madhan M

Department: Computer Science & Engineering | **Batch:** 2022-2026

⚠ THE PROBLEM

The Growing Threat of Credit Card Fraud

Credit card fraud has surged with the rapid growth of online transactions, mobile banking, and digital payment systems. Traditional detection methods rely on rule-based approaches or backend machine learning models that identify fraud only *after* transactions complete.

This reactive approach creates significant problems: financial losses accumulate, customer trust erodes, and fraudulent transactions slip through undetected until it's too late.

> Detection delays result in financial losses and reduced customer confidence in digital payment systems.



OUR SOLUTION

Real-Time Protection Through Intelligence

Dual Biometric Security

Fingerprint and facial recognition with liveness detection ensure only authorized cardholders approve transactions.

Machine Learning Detection

Real-time fraud prediction analyzes patterns and calculates dynamic risk scores before transactions complete.

Merchant Intelligence

Credibility analysis and transaction risk assessment evaluate merchant trustworthiness instantly.

Behavior Analysis

User behavior patterns and OTP-based alerts provide multi-level authentication for high-risk transactions.



How It Works: Prevention Before Loss



Transaction Initiated

Risk Score Calculated

Authentication Required

Approved or Blocked

Real-Time Prevention

Unlike traditional systems that detect fraud after completion, our application prevents fraudulent transactions *before* they occur.

01

Instant Analysis

Every transaction triggers immediate risk assessment combining ML models, merchant data, and user patterns.

02

Smart Authentication

High-risk transactions require dual biometric verification with facial liveness detection.

03

Seamless Experience

Low-risk transactions proceed smoothly while maintaining robust security protocols.

Impact and Benefits



Enhanced Security

Multi-layer authentication and real-time detection effectively reduce fraudulent transactions while preventing unauthorized access.



Smooth User Experience

Intelligent risk scoring ensures legitimate transactions flow seamlessly while suspicious activity triggers appropriate safeguards.



Cost-Effective Deployment

Secure, scalable system suitable for real-world implementation in modern digital banking environments without excessive infrastructure costs.



Restored Trust

Proactive fraud prevention rebuilds customer confidence in digital payment systems and strengthens banking relationships.

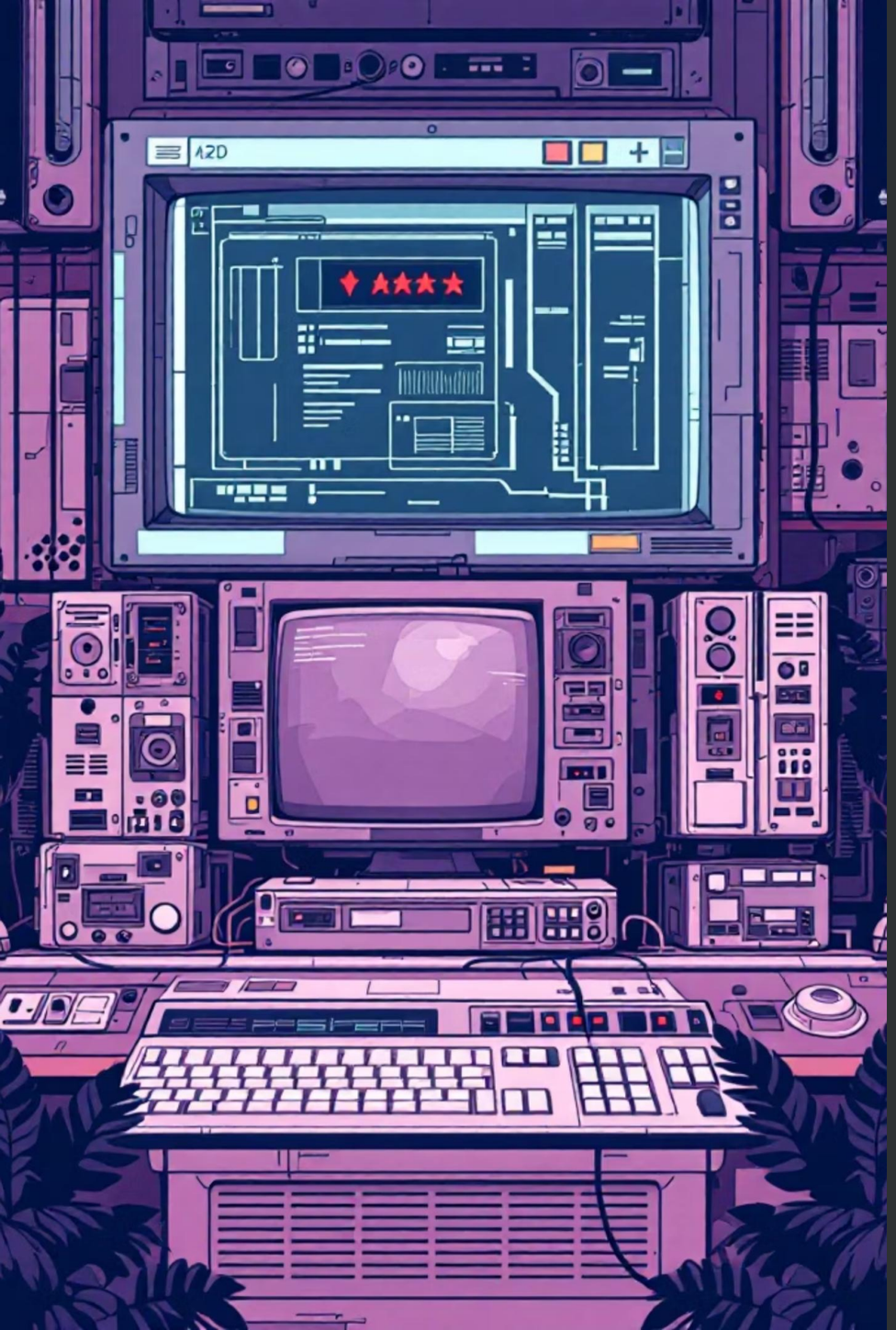
CHALLENGE

The Growing Threat of Credit Card Fraud



Rapid expansion of online transactions and digital payments has dramatically increased fraud risk. Fraudsters exploit stolen card details for unauthorized transactions, causing severe financial losses and eroding customer trust.

Traditional systems detect fraud only after completion, resulting in irreversible damage and reduced confidence in financial institutions.



Limitations of Existing Systems

Static Rule-Based Methods

Rely on historical data without real-time contextual analysis of user behavior or device information.

No Multi-Level Verification

Lack biometric authentication, OTP confirmation, and liveness detection mechanisms.

Delayed Detection

Binary outputs without adaptive security measures lead to higher false positives and post-transaction detection.

Feasibility Analysis

Technical Feasibility

Utilizes established Python ML libraries, Android biometric sensors, and cloud APIs for stable, scalable implementation.

Behavioral Feasibility

Users already familiar with fingerprint, face recognition, and OTP verification through existing banking apps.

Economic Feasibility

Relies on open-source tools and existing mobile hardware—no expensive infrastructure required.

SOLUTION

Proposed System Overview

A secure mobile-based application integrating machine learning, merchant intelligence, dual biometric authentication, OTP alerts, and facial liveness detection.



Dual Biometric Authentication

Fingerprint and face recognition for secure user verification



ML Risk Scoring

Dynamic risk assessment using ensemble machine learning models



Merchant Intelligence

Analyzes merchant credibility and fraud history



Real-Time Alerts

OTP verification and instant fraud notifications

System Requirements

Hardware Requirements

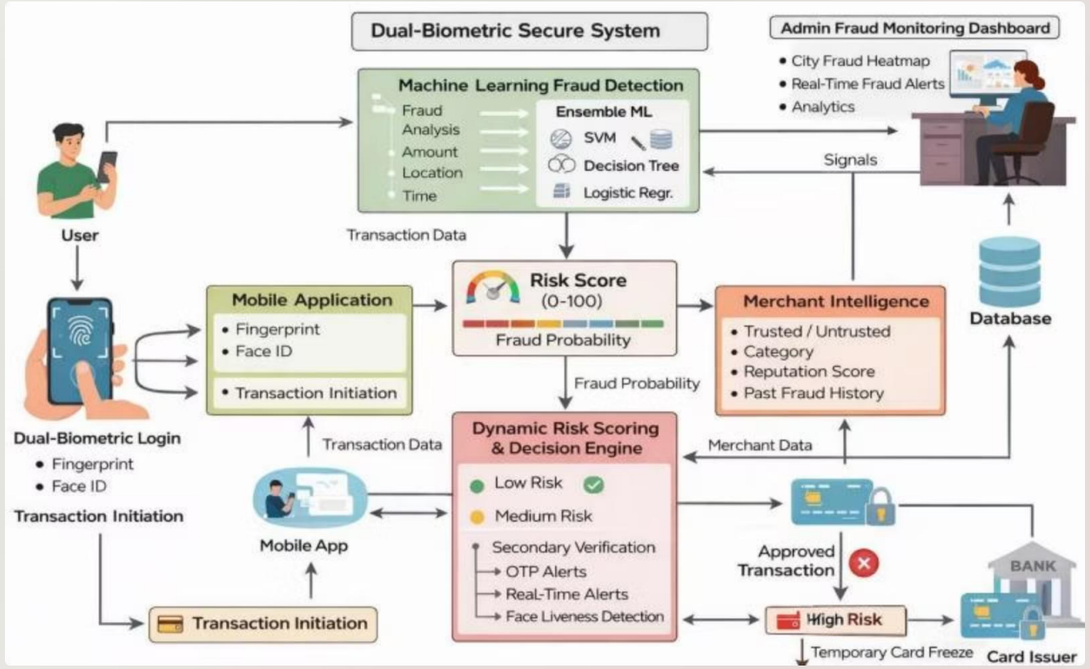
- Android Smartphone
- Fingerprint Sensor
- Front Camera
- Minimum 4 GB RAM
- Internet Connection

Software Requirements

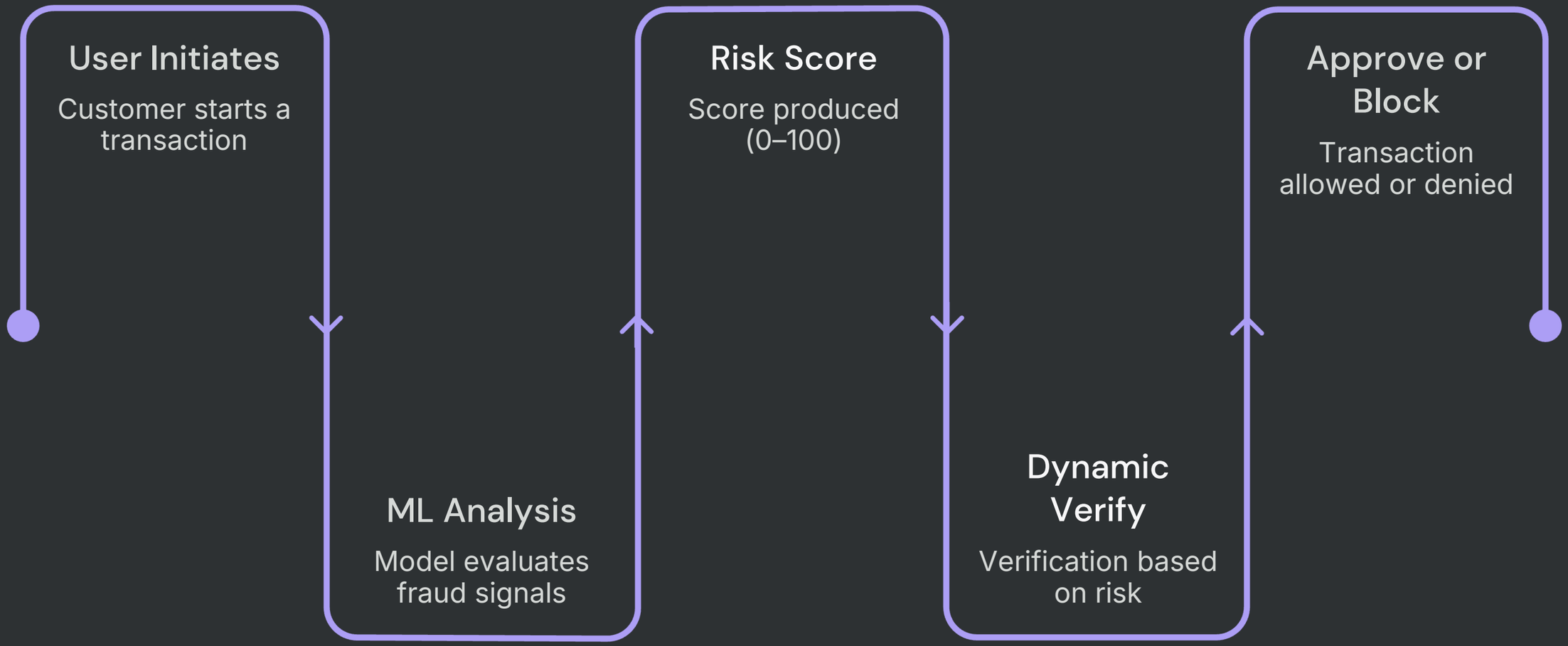
- OS: Android / Windows
- Languages: Python, Java / Kotlin
- ML Libraries: Scikit-learn
- Database: SQLite / Firebase
- Tools: Android Studio, VS Code

System Architecture

The architecture integrates mobile application, machine learning fraud detection, dynamic risk scoring, merchant intelligence, and admin monitoring dashboard for comprehensive fraud prevention.



Transaction Flow & Risk Assessment



Every transaction receives a dynamic risk score that determines the level of verification required, from low-risk approval to high-risk temporary card freeze.

Risk-Based Verification Levels

Low Risk

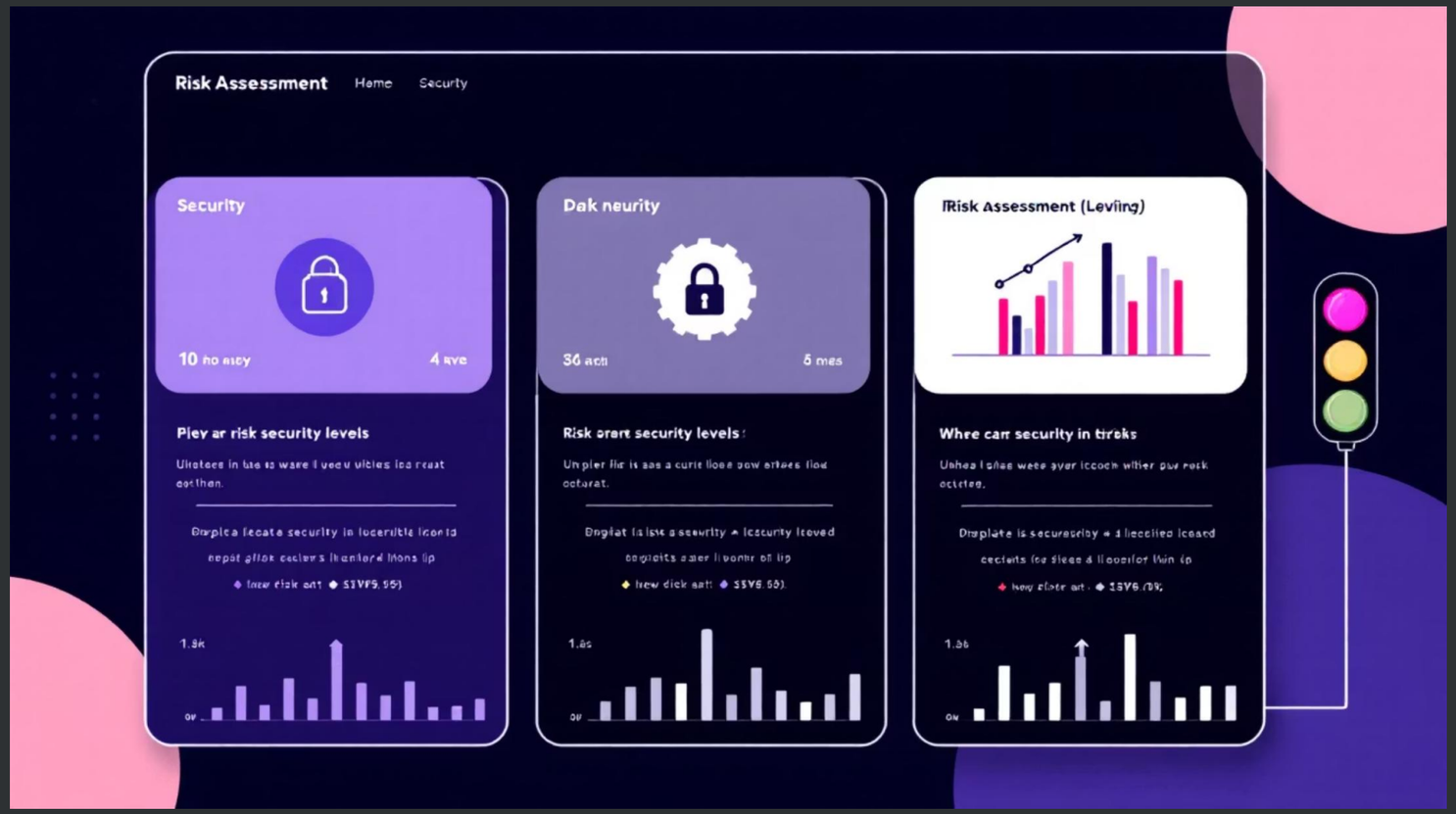
Instant approval with standard authentication

Medium Risk

OTP verification and real-time alerts required

High Risk

Face liveness detection and temporary card freeze



Key Benefits



Proactive Prevention

Stops fraud before transaction completion through real-time risk evaluation and multi-level verification.



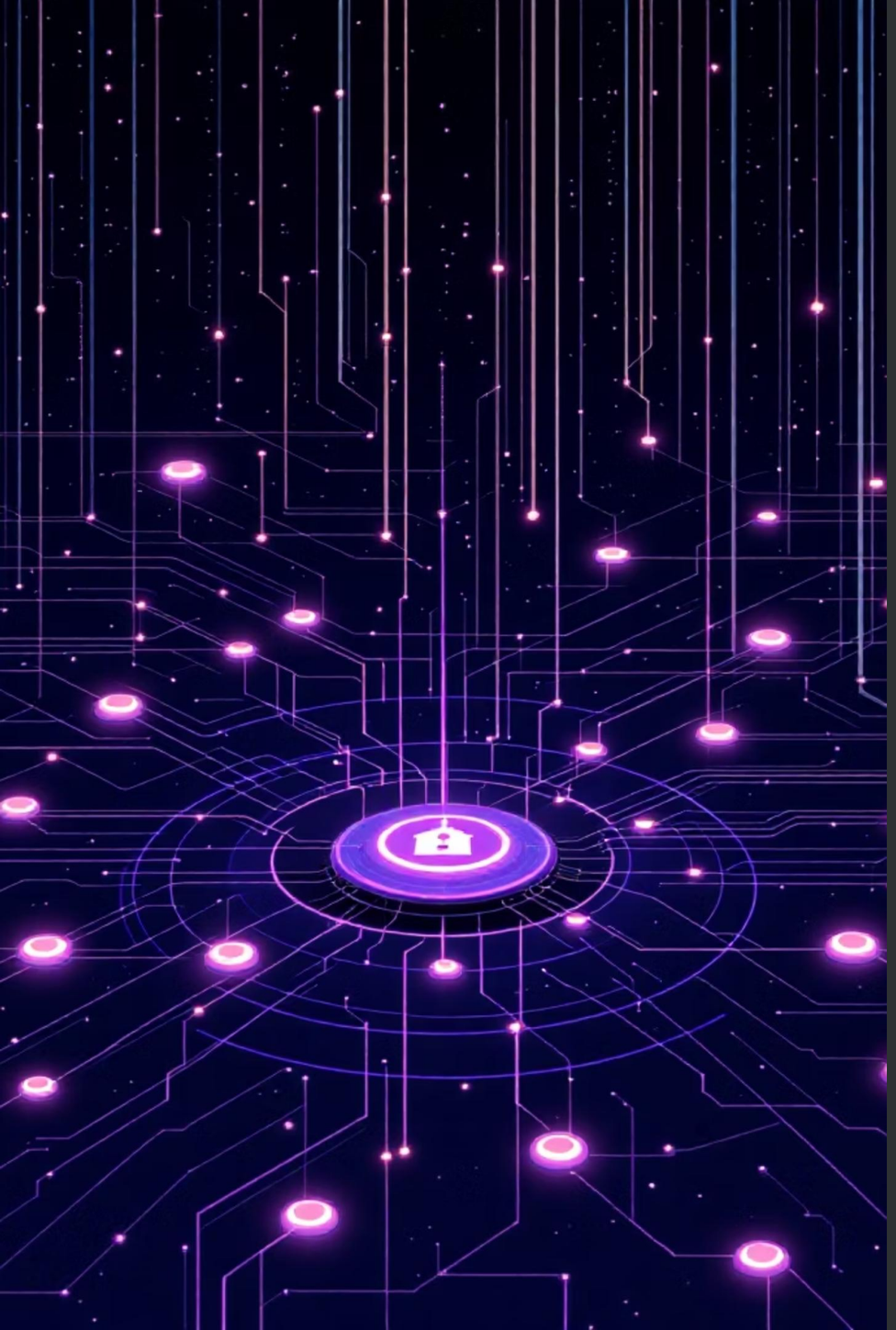
Reduced False Positives

Contextual analysis of user behavior, device data, and merchant credibility improves accuracy.



Enhanced Trust

Protects customers and financial institutions from losses while building confidence in digital payments.



Fraud Detection System

A comprehensive five-layer architecture combining machine learning, adaptive authentication, and real-time risk assessment to protect digital transactions.

ARCHITECTURE

System Architecture



01

Mobile Application

User login and transaction initiation

02

Backend API

Secure REST API communication

03

ML Model

Fraud probability prediction

04

Risk Scoring

Standardized 0-100 risk score

05

Decision Engine

Adaptive authentication rules

06

Database & Dashboard

Storage and fraud monitoring

MODULES

Core System Modules



Authentication

Verifies mobile number, fingerprint, and face ID

Transaction

Captures amount, merchant, location, and device data

Risk Engine

Calculates risk score using ML fraud probability × 100

Admin Panel

Displays fraud analytics and high-risk reports

DATABASE

Entity-Relationship Structure



Key Entities

- **User:** user_id, name, mobile, email
- **Card:** card_id, user_id, card_number, status
- **Transaction:** txn_id, card_id, amount, merchant_id, risk_score, status
- **Merchant:** merchant_id, trust_score
- **BlockStatus:** block_id, user_id, block_start, block_end

A User owns Cards that perform Transactions linked to Merchants, generating Risk Scores.

PROCESS

Transaction Flow



1

Login

Biometric authentication

2

Initiate

User starts transaction

3

Analyze

ML predicts fraud probability

4

Score

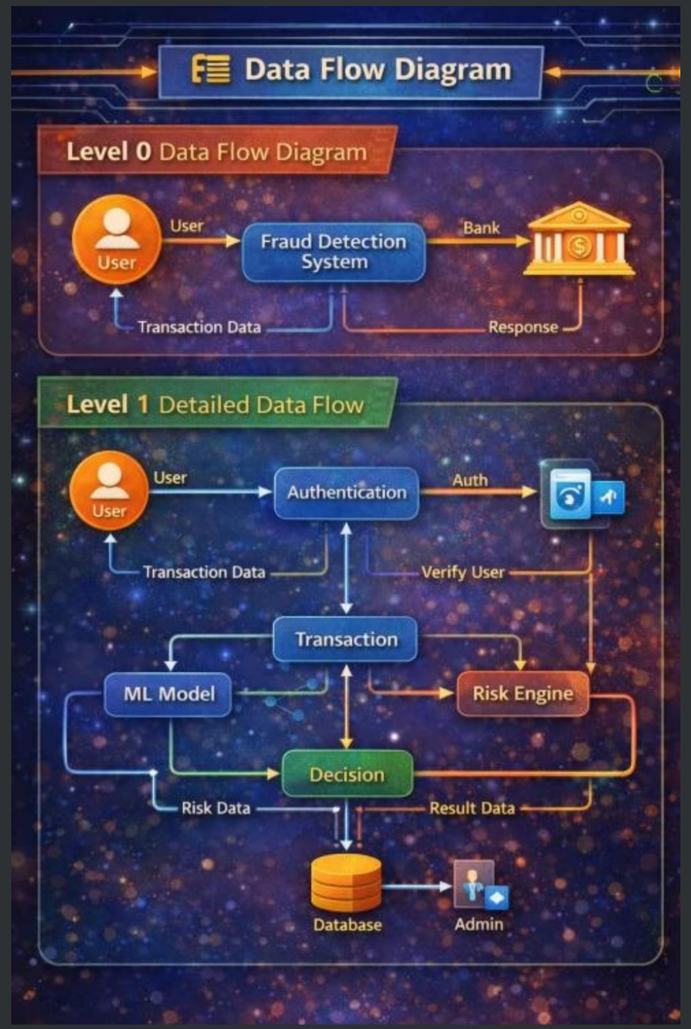
Risk score generated (0-100)

5

Decide

Approve or block transaction

Data Flow Architecture

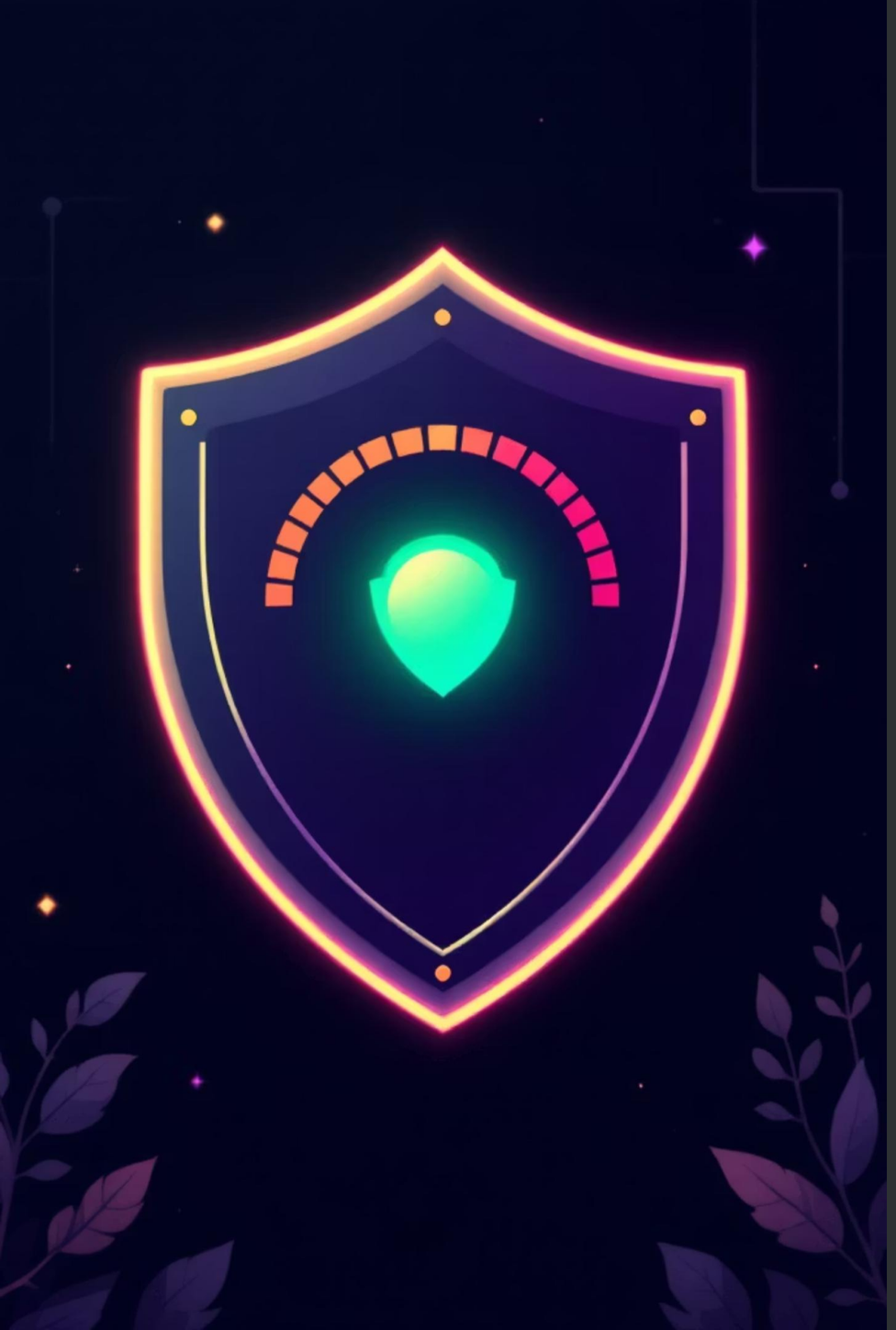


Two-Level DFD

Level 0: User → Fraud Detection System → Bank

Level 1: User → Authentication → Transaction → ML Model → Risk Engine → Decision Engine → Database → Admin Dashboard

Data flows independently across components, separating prediction logic from security enforcement.



SECURITY

Adaptive Authentication Thresholds

Risk < 50	50 ≤ Risk < 70	70 ≤ Risk < 90
Fingerprint + Face ID	Fingerprint + Face ID + OTP	OTP + Live Face Detection

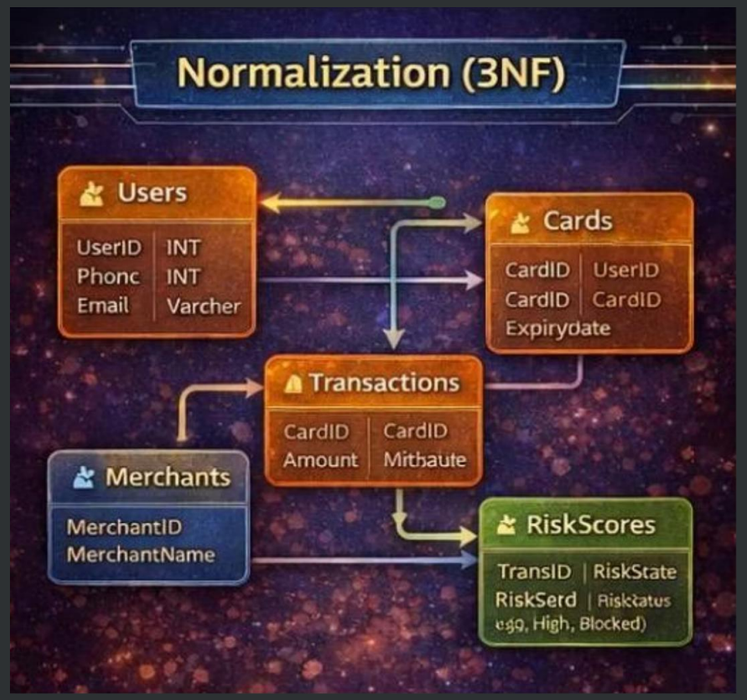
Risk ≥ 90

SMS Alert + 6-hour block if no response

Threshold-based logic applies progressively stronger authentication as risk increases, with merchant trust ratings adjusting final scores.

NORMALIZATION

Database Efficiency



1

First Normal Form

Atomic values without repeating groups

2

Second Normal Form

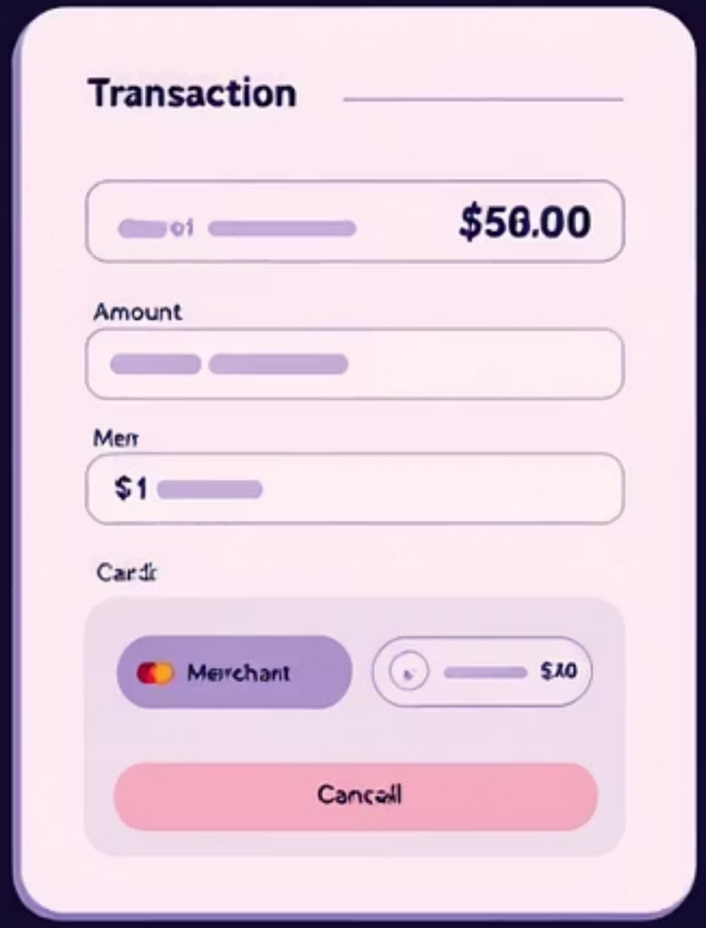
Non-key attributes fully depend on primary key

3

Third Normal Form

Removes transitive dependency

This structure prevents redundancy and maintains data integrity across all tables.



USE CASES

System Actors & Actions

User Actions

- Login with biometrics
- Make transaction
- Verify OTP
- Perform face detection

Admin Actions

- Monitor fraud activity
- View analytics reports
- Manage blocked accounts

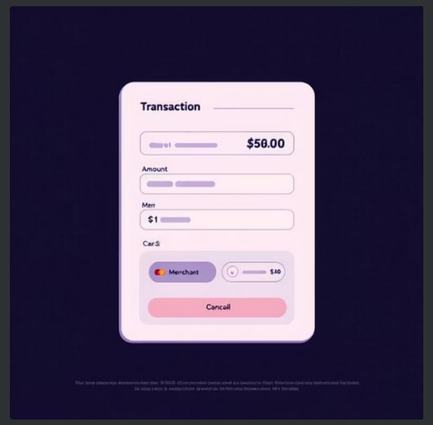
INTERFACE

Form Design Principles



Login Form

Mobile number + biometric authentication



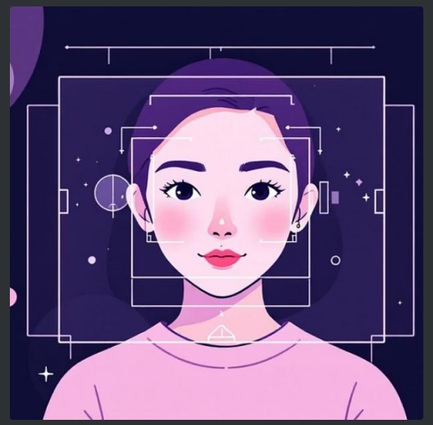
Transaction Form

Amount, merchant, location auto-detection



OTP Verification

Secure code confirmation with resend option



Live Face Detection

Camera-based identity confirmation



Admin Dashboard

Fraud analytics and account monitoring

Simple, user-friendly interfaces with minimal input fields, strong validation, real-time feedback, and clear error messages.